MANUFACTURING COST AND PROFIT ANALYSIS

Python Programming Case Study

1. Problem Statement

Manufacturing businesses need to understand the total cost of producing goods and whether the selling price generates the expected profit. Calculating material, labour, wastage, and machine costs manually can lead to mistakes. A simple program can combine these costs, calculate profit and profit percentage, and indicate whether the profit target has been achieved.

2. Objective

To develop a Python program that accepts production expenses and selling price, calculates total production cost, profit, and profit percentage, displays a production report, and checks the result against a user-defined profit target.

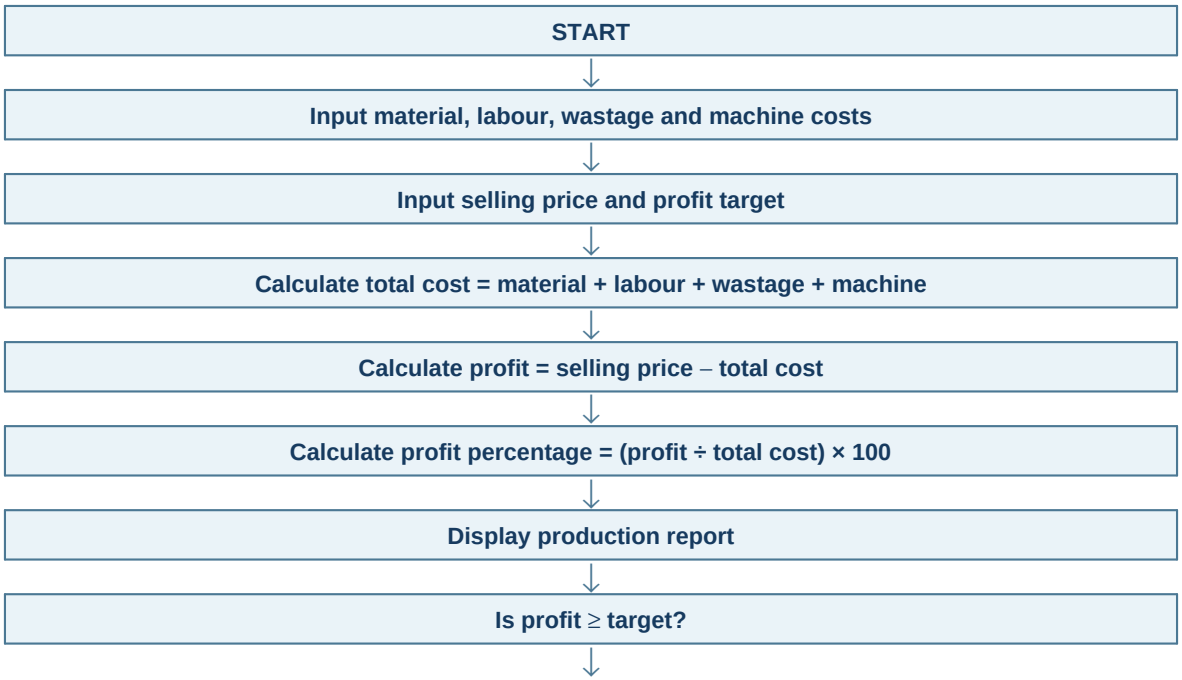
3. Assumptions

- All cost, selling-price, and target inputs are entered as valid non-negative numeric values.
- Material, labour, wastage, and machine costs are the only production-cost components considered.
- Total production cost is the sum of these four cost components.
- Profit is calculated as selling price minus total production cost.
- Profit percentage is calculated as (profit ÷ total cost) × 100; total cost must be greater than zero.
- The target is considered achieved when calculated profit is greater than or equal to the entered target.

4. Algorithm

1. Start the program.
2. Read material cost, labour charges, wastage cost, machine cost, selling price, and profit target.
3. Calculate total production cost by adding material, labour, wastage, and machine costs.
4. Calculate profit as selling price minus total production cost.
5. Calculate profit percentage using (profit / total cost) × 100.
6. Display total cost, selling price, profit, and profit percentage.
7. Compare profit with the target.
8. Display “Target Achieved” if profit is greater than or equal to target; otherwise display “Target Not Achieved”.
9. Stop the program.

5. Flowchart




Yes: Display “Target Achieved” / No: Display “Target Not Achieved”

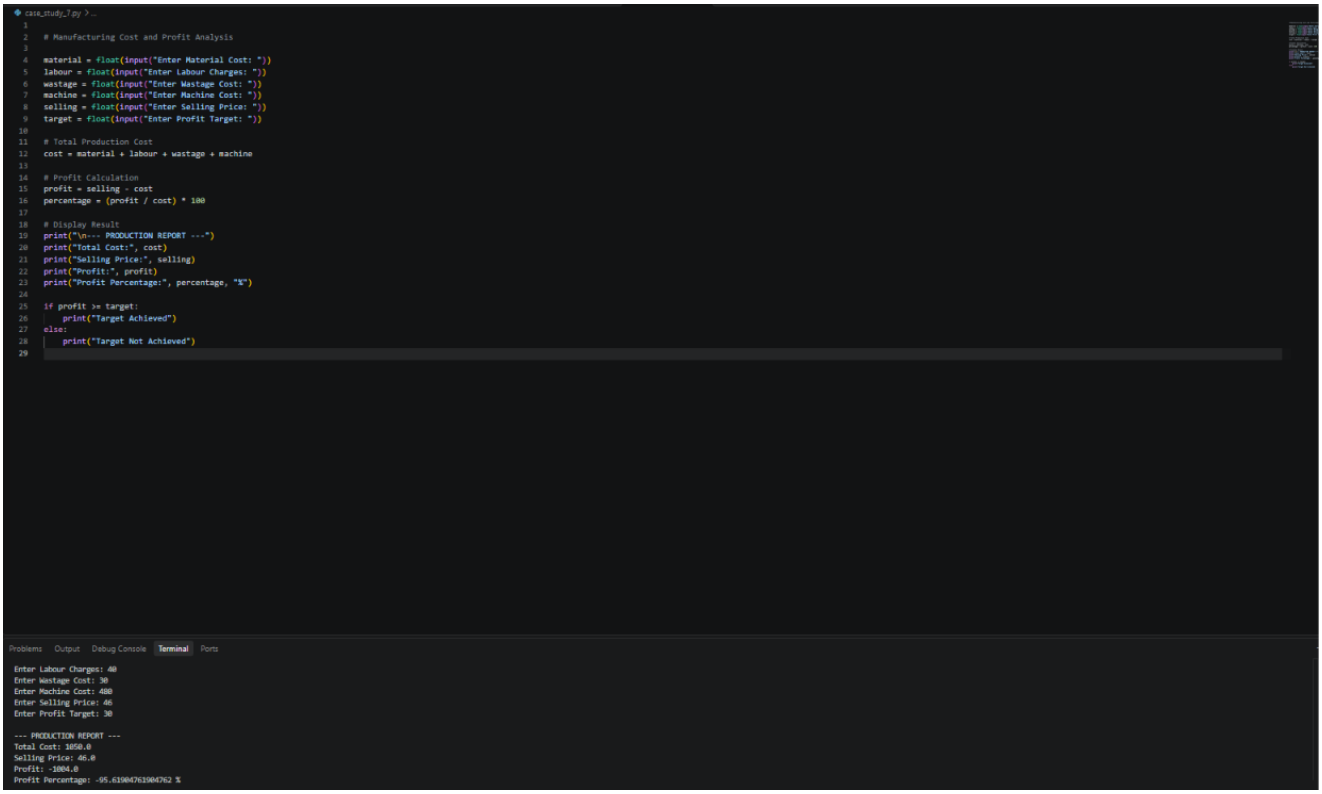
STOP

6. Pseudocode

```
BEGIN
INPUT material, labour, wastage, machine
INPUT selling_price, target
cost = material + labour + wastage + machine
profit = selling_price - cost
percentage = (profit / cost) * 100
DISPLAY cost, selling_price, profit, percentage
IF profit >= target THEN
DISPLAY 'Target Achieved'
ELSE
DISPLAY 'Target Not Achieved'
END IF
END
```

7. Python Source Code and Output Screenshot

The original screenshot provided is included on the next page.



8. Conclusion

The Manufacturing Cost and Profit Analysis program combines major production expenses, calculates profit and profit percentage, and checks whether the expected profit target is met. It demonstrates the use of Python input, arithmetic operations, formatted output, and conditional statements in a practical manufacturing-cost scenario.